INCH-POUND

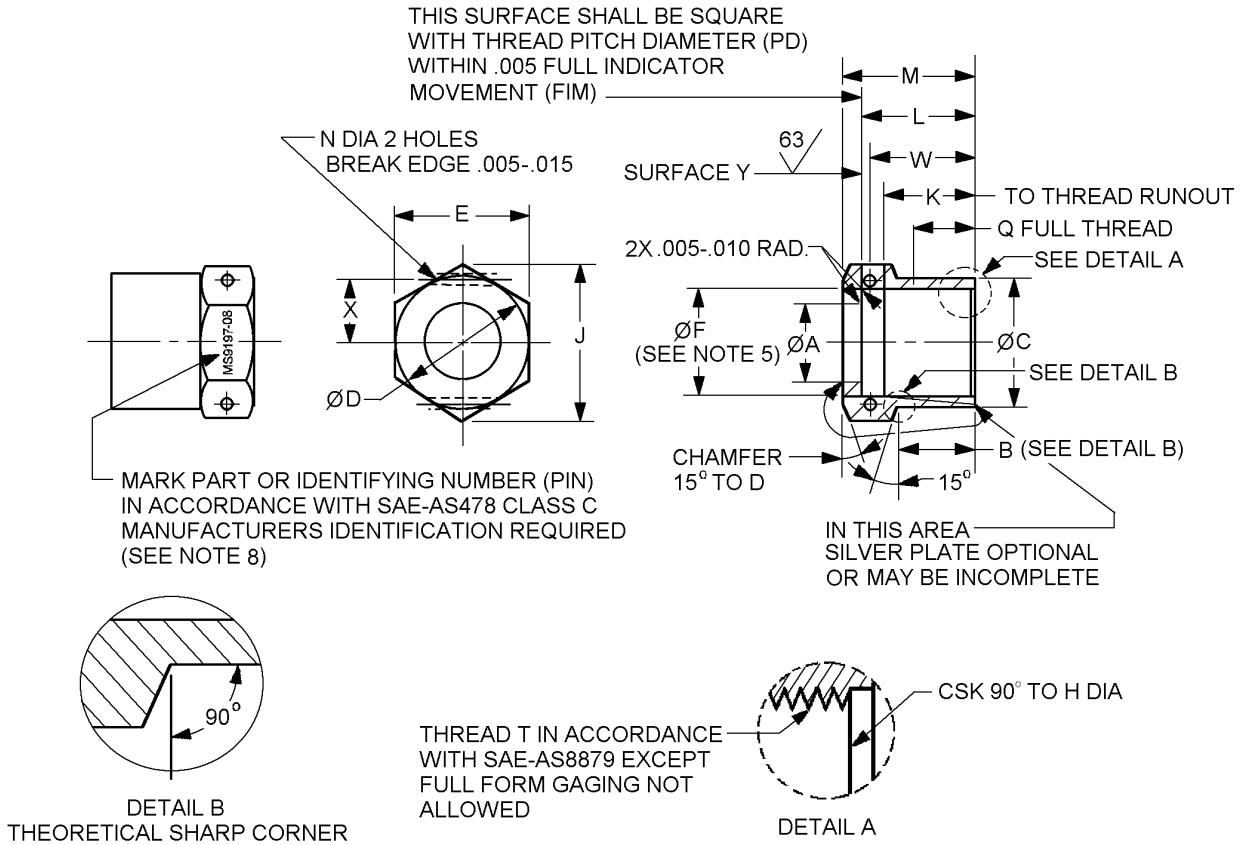
MS9197D
16 February 2010
SUPERSEDING
MS9197C
13 December 1972

DETAIL SPECIFICATION SHEET

NUT, TUBE COUPLING-CRES SAE-AMS5646, SILVER PLATE

This specification is approved for use by all Departments and Agencies of the Department of Defense .

The requirements for acquiring the product described herein shall consist of this specification sheet.



Inches	mm
.005	0.13
.010	0.25
.015	0.38

FIGURE 1. Nut, tube coupling.

AMSC N/A

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:12Z
Check the source to verify that this is the current version before use.

Dash number	Tube OD inches (mm)	Thread T SAE-AS8879	A diameter inches (mm) +.003 (0.08) -.000	B inches (mm) ±.015 (0.38)	C diameter inches (mm) ±.005 (0.13)
-02	.125 (3.18)	.3125 - 24UNJF -3B	.180 (4.57)	.435 (11.05)	.425 (10.80)
-03	.188 (4.78)	.3750 - 24UNJF -3B	.242 (6.15)	.452 (11.48)	.490 (12.45)
-04	.250 (6.35)	.4375 - 20UNJF -3B	.305 (7.75)	.467 (11.86)	.550 (13.97)
-05	.313 (7.95)	.5000 - 20UNJF -3B	.374 (9.50)	.440 (11.18)	.615 (15.62)
-06	.375 (9.53)	.5625 - 18UNJF - 3B	.440 (11.18)	.467 (11.86)	.678 (17.22)
-07	.438 (11.13)	.6250 - 18UNJF -3B	.502 (12.75)	.493 (12.52)	.740 (18.80)
-08	.500 (12.70)	.7500 - 16UNJF - 3B	.570 (14.48)	.493 (12.52)	.865 (21.97)
-09	.562 (14.27)	.8125 - 16UNJ - 3B	.634 (16.10)	.493 (12.52)	.927 (23.55)
-10	.625 (15.88)	.8750 - 14UNJF - 3B	.698 (17.73)	.617 (15.67)	.990 (25.15)
-11	.688 (17.48)	1.000 - 12UNJF - 3B	.770 (19.56)	.588 (14.94)	1.115 (28.32)
-12	.750 (19.05)	1.0625 - 12UNJ - 3B	.834 (21.18)	.588 (14.94)	1.180 (29.97)
-14	.875 (22.23)	1.1875 - 12UNJ - 3B	.961 (24.41)	.615 (15.62)	1.320 (33.53)
-16	1.000 (25.40)	1.3125 - 12UNJ - 3B	1.089 (27.66)	.652 (16.56)	1.433 (36.40)
-18	1.125 (28.58)	1.500 - 12UNJF - 3B	1.217 (30.91)	.680 (17.27)	1.620 (41.15)
-20	1.250 (31.75)	1.6250 - 12UNJ - 3B	1.347 (34.21)	.727 (18.47)	1.745 (44.32)
-24	1.500 (38.10)	1.8750 - 12UNJ - 3B	1.617 (41.07)	.880 (22.35)	1.995 (50.67)
-28	1.750 (44.45)	2.2500 - 12UNJ - 3B	1.890 (48.01)	.930 (23.62)	2.410 (61.21)
-32	2.000 (50.80)	2.500 - 12UNJ - 3B	2.167 (55.04)	1.060 (26.92)	2.660 (67.56)
-40	2.500 (63.50)	3.000 - 12UNJ - 3B	2.667 (67.74)	.860 (21.84)	3.200 (81.28)
-48	3.000 (76.20)	3.500 - 12UNJ - 3B	3.180 (80.77)	.910 (23.11)	3.700 (93.98)

FIGURE 1. Nut, tube coupling - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:12Z
Check the source to verify that this is the current version before use.

Dash number	D diameter inches (mm)	E inches (mm)	F diameter inches (mm)	H inches (mm) +.030 (0.76) -.000
-02	.438 (11.13)	.430 - .439 (10.92) - (11.15)	.272 - .280 (6.91) - (7.11)	.312 (7.92)
-03	.500 (12.70)	.492 - .502 (12.50) - (12.75)	.334 - .342 (8.48) - (8.69)	.375 (9.53)
-04	.562 (14.27)	.553 - .564 (14.05) - (14.33)	.388 - .397 (9.86) - (10.08)	.438 (11.13)
-05	.625 (15.88)	.616 - .627 (15.65) - (15.93)	.451 - .459 (11.46) - (11.66)	.500 (12.70)
-06	.688 (17.48)	.679 - .690 (17.25) - (17.53)	.508 - .517 (12.90) - (13.13)	.562 (14.27)
-07	.750 (19.05)	.741 - .752 (18.82) - (19.10)	.570 - .579 (14.48) - (14.71)	.625 (15.88)
-08	.875 (22.23)	.865 - .877 (21.97) - (22.28)	.689 - .698 (17.50) - (17.73)	.750 (19.05)
-09	.937 (23.80)	.928 - .940 (23.57) - (23.88)	.751 - .760 (19.08) - (19.30)	.812 (20.62)
-10	1.000 (25.40)	.990 - 1.002 (25.15) - (25.45)	.805 - .815 (20.45) - (20.70)	.875 (22.23)
-11	1.125 (28.58)	1.114 - 1.127 (28.30) - (28.63)	.918 - .930 (23.32) - (23.62)	1.000 (25.40)
-12	1.250 (31.75)	1.239 - 1.252 (31.47) - (31.80)	.981 - .991 (24.92) - (25.17)	1.062 (26.97)
-14	1.375 (34.93)	1.364 - 1.377 (34.65) - (34.98)	1.106 - 1.116 (28.09) - (28.35)	1.188 (30.18)
-16	1.500 (38.10)	1.489 - 1.502 (37.82) - (38.15)	1.231 - 1.241 (31.27) - (31.52)	1.312 (33.32)
-18	1.750 (44.45)	1.738 - 1.752 (44.15) - (44.50)	1.419 - 1.429 (36.04) - (36.30)	1.500 (38.10)
-20	1.812 (46.02)	1.800 - 1.814 (45.72) - (46.08)	1.544 - 1.554 (39.22) - (39.47)	1.625 (41.28)
-24	2.125 (53.98)	2.113 - 2.127 (53.67) - (54.03)	1.794 - 1.804 (45.57) - (45.82)	1.875 (47.63)
-28	2.500 (63.50)	2.487 - 2.502 (63.17) - (63.55)	2.169 - 2.179 (55.09) - (55.35)	2.250 (57.15)
-32	2.750 (69.85)	2.737 - 2.752 (69.52) - (69.90)	2.419 - 2.429 (61.44) - (61.70)	2.500 (63.50)
-40	3.250 (82.55)	3.237 - 3.252 (82.22) - (82.60)	2.919 - 2.929 (74.14) - (74.40)	3.000 (76.20)
-48	3.812 (96.82)	3.798 - 3.814 (96.47) - (96.88)	3.419 - 3.429 (86.84) - (87.10)	3.500 (88.90)

FIGURE 1. Nut, tube coupling - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:12Z
Check the source to verify that this is the current version before use.

Dash number	J min	K max inches (mm)	L inches (mm) ±.005 (0.13)	M inches (mm)	N inches (mm)
-02	.491 (12.47)	.425 (10.80)	.575 (14.61)	.650 (16.51)	.045 (1.14) - .050 (1.27)
-03	.561 (14.25)	.441 (11.20)	.592 (15.04)	.667 (16.94)	.057 (1.45) - .062 (1.57)
-04	.631 (16.03)	.482 (12.24)	.607 (15.42)	.682 (17.32)	
-05	.703 (17.86)	.482 (12.24)	.610 (15.49)	.690 (17.53)	.065 (1.65) - .075 (1.91)
-06	.775 (19.69)	.535 (13.59)	.637 (16.18)	.742 (18.85)	
-07	.846 (21.49)	.536 (13.61)	.663 (16.84)	.768 (19.51)	
-08	.987 (25.07)	.569 (14.45)	.723 (18.36)	.828 (21.03)	
-09	1.059 (26.90)	.596 (15.14)	.723 (18.36)	.828 (21.03)	
-10	1.130 (28.70)	.667 (16.94)	.817 (20.75)	.927 (23.55)	
-11	1.271 (32.28)	.738 (18.75)	.868 (22.05)	.968 (24.59)	
-12	1.414 (35.92)	.738 (18.75)	.868 (22.05)	.968 (24.59)	
-14	1.556 (39.52)	.765 (19.43)	.915 (23.24)	1.015 (25.78)	
-16	1.699 (43.15)	.786 (19.96)	.962 (24.43)	1.062 (26.97)	
-18	1.985 (50.42)	.800 (20.32)	.990 (25.15)	1.120 (28.45)	
-20	2.054 (52.17)	.810 (20.57)	1.017 (25.83)	1.167 (29.64)	
-24	2.411 (61.24)	.937 (23.80)	1.170 (29.72)	1.320 (33.53)	
-28	2.838 (72.09)	.945 (24.00)	1.170 (29.72)	1.370 (34.80)	
-32	3.123 (79.32)	1.050 (26.67)	1.300 (33.02)	1.500 (38.10)	
-40	3.693 (93.80)	.850 (21.59)	1.100 (27.94)	1.300 (33.02)	
-48	4.334 (110.08)	.900 (22.86)	1.150 (29.21)	1.350 (34.29)	

FIGURE 1. Nut, tube coupling - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:12Z
Check the source to verify that this is the current version before use.

Dash number	Q min	W max inches (mm)	X inches (mm) ±.005 (0.13)
-02	.300 (7.62)	.535 (13.59)	.180 (4.57) - .190 (4.83)
-03	.316 (8.03)	.552 (14.02)	.210 (5.33) - .220 (5.59)
-04	.332 (8.43)	.567 (14.40)	.245 (6.22) - .255 (6.48)
-05	.332 (8.43)	.565 (14.35)	.280 (7.11) - .290 (7.37)
-06	.367 (9.32)	.592 (15.04)	.310 (7.87) - .320 (8.13)
-07	.368 (9.35)	.618 (15.70)	.336 (8.53) - .356 (9.04)
-08	.383 (9.73)	.648 (16.46)	.400 (10.16) - .420 (10.67)
-09	.383 (9.73)	.668 (16.97)	.427 (10.85) - .447 (11.35)
-10	.453 (11.51)	.777 (19.74)	.465 (11.81) - .485 (12.32)
-11	.488 (12.40)	.818 (20.78)	.520 (13.21) - .540 (13.72)
-12	.488 (12.40)	.818 (20.78)	.580 (14.73) - .600 (15.24)
-14	.515 (13.08)	.865 (21.97)	.645 (16.38) - .665 (16.89)
-16	.536 (13.61)	.912 (23.16)	.715 (18.16) - .735 (18.67)
-18	.550 (13.97)	.945 (24.00)	.840 (21.34) - .860 (21.84)
-20	.560 (14.22)	.967 (24.56)	.890 (22.61) - .910 (23.11)
-24	.687 (17.45)	1.100 (27.94)	1.040 (26.42) - 1.060 (26.92)
-28	.695 (17.65)	1.170 (29.72)	1.255 (31.88) - 1.275 (32.39)
-32	.800 (20.32)	1.300 (33.02)	1.395 (35.43) - 1.415 (35.94)
-40	.600 (15.24)	1.100 (27.94)	1.680 (42.67) - 1.700 (43.18)
-48	.650 (16.51)	1.150 (29.21)	2.000 (50.80) - 2.020 (51.31)

FIGURE 1. Nut, tube coupling - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:12Z
Check the source to verify that this is the current version before use.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Tolerances linear dimensions ±.010 inch (0.25 mm) angular dimensions ±5°.
4. Surface texture unless otherwise specified shall be 125µ inches (3.17µm) except hex, reference ASME B46.1.
5. Diameters A and F shall be concentric to thread PD within .004 inch (0.01 mm) FIM. Diameter C and hexagon E shall be concentric to thread PD within .010 inch (0.25 mm) FIM.
6. Dimensions specified are after plating.
7. Break Sharp edges .003 - .015 inch (0.08 - 0.38 mm) unless otherwise specified. Fillets .005 - .020 inch (0.13 - 0.51 m) RAD or chamfer
8. Multiple marking permissible provided positive identification is shown. Parts shall be marked before threading.
9. For design features purposes, this standard takes precedence over documents referenced herein.

FIGURE 1. Nut, tube coupling - Continued.

REQUIREMENTS:

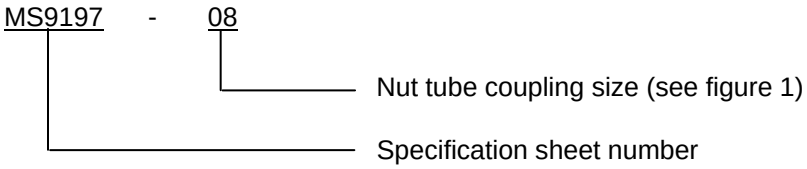
Dimensions and configurations: See figure 1.

Material: Steel, corrosion and heat-resistant in accordance with SAE-AMS5646.

Surface finish: Silver plate in accordance with SAE-AMS2410, 300µ to 600µ inches (7.62 µm to 15.24µm) thick on thread at PD and surface Y.

Parts shall be fluorescent penetrate inspected in accordance with ASTM E1417.

PIN:



PIN Example:

MS9197-08 identifies a nut tube coupling .500 inch (12.70 mm) tube.

Do not use unassigned PIN's.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:12Z
Check the source to verify that this is the current version before use.

The approximate weight for each fitting is shown in table I

TABLE I. Approximate weight.

Dash number	Approximate weight lb/100 (kg/100)		Dash number	Approximate weight lb/100 (kg/100)
-02	1.210 (0.55)		-11	7.859 (3.56)
-03	1.475 (0.67)		-12	9.907 (4.49)
-04	1.830 (0.83)		-14	12.164 (5.52)
-05	2.600 (1.18)		-16	13.637 (6.19)
-06	2.920 (1.32)		-18	19.830 (8.99)
-07	3.422 (1.55)		-20	20.016 (9.08)
-08	4.680 (2.12)		-24	27.903 (12.66)
-09	5.314 (2.41)		-28	41.710 (18.92)
-10	6.310 (2.86)		-32	49.000 (22.23)
			-40	60.621 (27.50)
			-48	78.529 (35.62)

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. This document references the following:

ASME B46.1
ASTM E1417
SAE-AS478
SAE-AS8879
SAE-AMS2410
SAE-AMS5646

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2009-083)

Review activities:
Navy - MC
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:12Z
Check the source to verify that this is the current version before use.